AI-Policy

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Background

Existing research has examined AI governance through various lenses; however, the relationship between economic freedom and AI-related policy discourse remains underexplored. This research seeks to address that gap by investigating whether a country's level of economic freedom shapes how governments frame the development and regulation of AI within national strategy documents.

In particular, this study compares the G7 countries and China to analyze how different economic and political environments influence AI regulatory narratives, policy priorities, and governance orientations.

Research Question

> How do variations in economic freedom and AI vibrancy shape the dominant framing embedded in national discourse, and how do these relationships vary across G7 countries and China?

Hypotheses

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H1 — AI Vibrancy and Innovation-Oriented Rhetoric

Countries with higher Stanford AI Vibrancy scores will exhibit more innovation-enabling rhetoric in their national policy documents.

Rationale

Countries with more developed AI ecosystems are expected to have stronger industry stakeholders and innovation-oriented policy agendas, encouraging governments to frame AI regulation in enabling rather than restrictive terms.

H2 — AI Vibrancy and Risk-Oriented Rhetoric

Countries with lower Stanford AI Vibrancy scores will exhibit more risk-enabling rhetoric in their national policy documents.

Rationale

Countries with less developed AI ecosystems are expected to have weaker industry stakeholders and risk-oriented policy agendas, encouraging governments to frame AI regulation in restrictive rather than enabling terms.

H3 — Economic Freedom and Positive Regulatory Framing

Countries whose AI regulatory discourse is more positive and innovation-oriented will exhibit higher economic freedom scores.

Rationale

Positive regulatory framing may reflect broader ideological commitments toward market liberalism and economic openness, which are captured by economic freedom indicators.

H4 — Economic Freedom and Negative Regulatory Framing

Countries whose AI regulatory discourse is more negative and risk-oriented will exhibit lower economic freedom scores.

Rationale

Negative regulatory framing may reflect broader ideological commitments away from market liberalism and economic openness, which are captured by economic freedom indicators.

Methodology

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AI Vibrancy

AI vibrancy is measured using the Stanford AI Index country-level AI Vibrancy scores for the G7 countries and China. The main country-level predictor is the **Total Score**, renamed in the analysis as `AI Vibrancy Score`. Component dimensions such as R&D, Responsible AI, Economy, Talent, Policy and Governance, Public Opinion, and Infrastructure are retained for descriptive analysis.

Economic Freedom

Economic freedom is measured using the Heritage Foundation Index of Economic Freedom panel. The correlation study uses the latest available year in the local panel, currently **2026**, and uses `Overall Score` as `IEF Overall Score`.

Text Analysis

National AI strategy and regulatory documents from G7 countries, China, and the European Union are analyzed using sentence-level AI regulatory framing. English documents are used directly. Chinese-heavy texts are translated into English with `facebook/nllb-200-distilled-600M` or analyzed through an existing English analysis corpus where available.

The current text-analysis workflow uses **DeepSeek R1 14B through Ollama** for local sentence classification. Each sentence is classified independently into one of five labels:

Label	Meaning
Innovation-oriented	Emphasizes innovation, investment, adoption, competitiveness, productivity, and flexible regulation
Risk-oriented	Emphasizes risk, safety, accountability, privacy, fairness, human rights, and regulatory obligations
Mixed	A single sentence contains both innovation and risk framing
Neutral	Descriptive, administrative, or procedural sentences
Unsure	Unclear residual category; excluded from the no-unsure human-gold calibration set

The model prompt includes the project codebook and human-annotated gold examples. The model returns structured JSON with a label, confidence score, and reason. Both the raw model label and the final corrected label are retained for auditing.

Human Calibration and Correction

The LLM labeling workflow is calibrated with human expert annotations rather than treated as a fully automatic black box. A sample of 40 sentences was independently annotated by multiple human annotators. Annotator 1 showed lower annotation stability, so the final calibration workflow uses the Daria + Stephen consensus labels.

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Daria and Stephen agreed on 23 sentences. For the current `daria_stephen_no_unsure_full` run, human-gold `Unsure` labels are excluded, leaving **16 substantive gold labels** for few-shot examples, exact sentence overrides, and model evaluation. Raw DeepSeek accuracy on this usable gold set is **75%**; after exact human-gold overrides, corrected accuracy is **100%**. The full run applies **4 human overrides**.

Correction logic:

```
corrected_label =
    human_gold_label, if the sentence is present in the usable human
gold set
    model_label, otherwise
```

Country/entity framing summaries are computed from corrected labels. Innovation-oriented sentences score `+1`, risk-oriented sentences score `-1`, and mixed, neutral, and unsure sentences score `0` for the net framing measure. The main aggregate variables are `Innovation Share`, `Risk Share`, `Mean Framing Score`, and the smoothed `Innovation-to-Risk Ratio`.

Correlation Analysis

Correlation analysis is conducted in `notebooks/Correlation Study/correlation_study.ipynb` to examine relationships between:

- Stanford AI Vibrancy scores and innovation-oriented framing
- Stanford AI Vibrancy scores and risk-oriented framing
- Economic Freedom scores and innovation-oriented framing
- Risk-oriented framing and Economic Freedom scores
- Combined AI Vibrancy + Economic Freedom predictors and the log innovation-to-risk framing ratio

The current notebook uses the corrected DeepSeek framing output by default: `outputs/deepseek_ai_framing_summary_corrected_daria_stephen_no_unsure_full.csv` . The European Union is included in descriptive framing summaries but excluded from the G7 + China correlation sample because comparable AI Vibrancy and IEF country scores are not used for it in this study.

Because the merged country-level sample has only eight observations, all correlation coefficients, regressions, and mediation diagnostics are interpreted as exploratory and directional rather than confirmatory.

Expected Contribution

This research contributes to the emerging literature on AI governance by connecting political-economic structure with regulatory discourse.

Rather than treating AI policy solely as a technical or legal issue, this study examines how broader economic ideology may shape national narratives surrounding AI development and regulation.

Core Analytical Logic

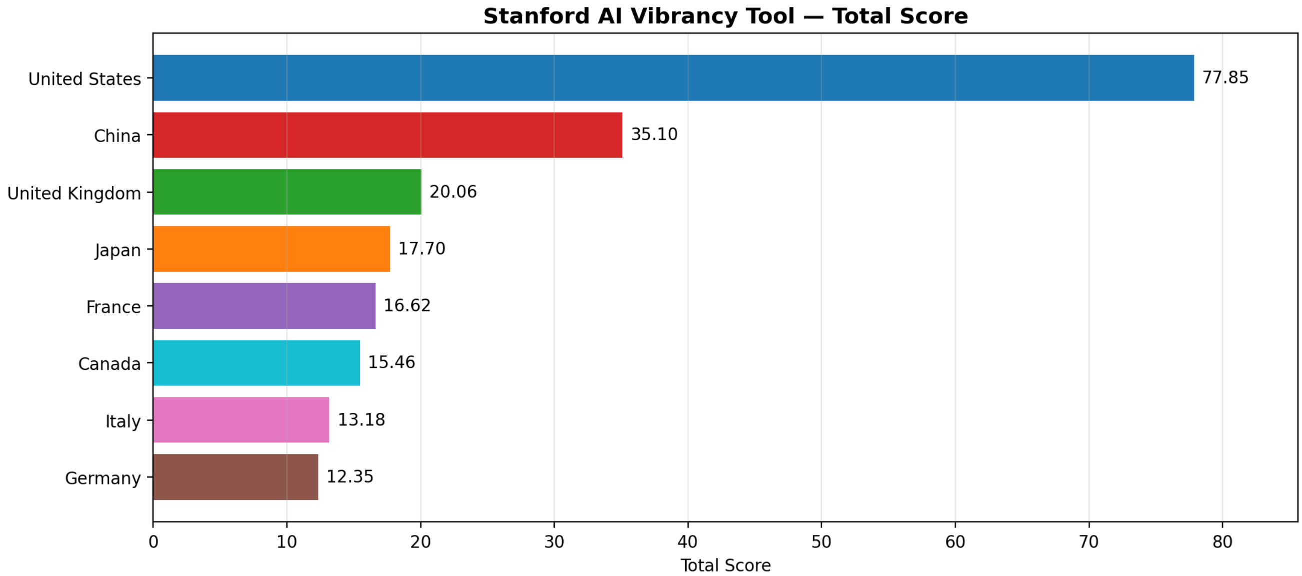
```
AI Vibrancy
      ↓
Regulatory Framing
      ↓
Economic Freedom Orientation
```

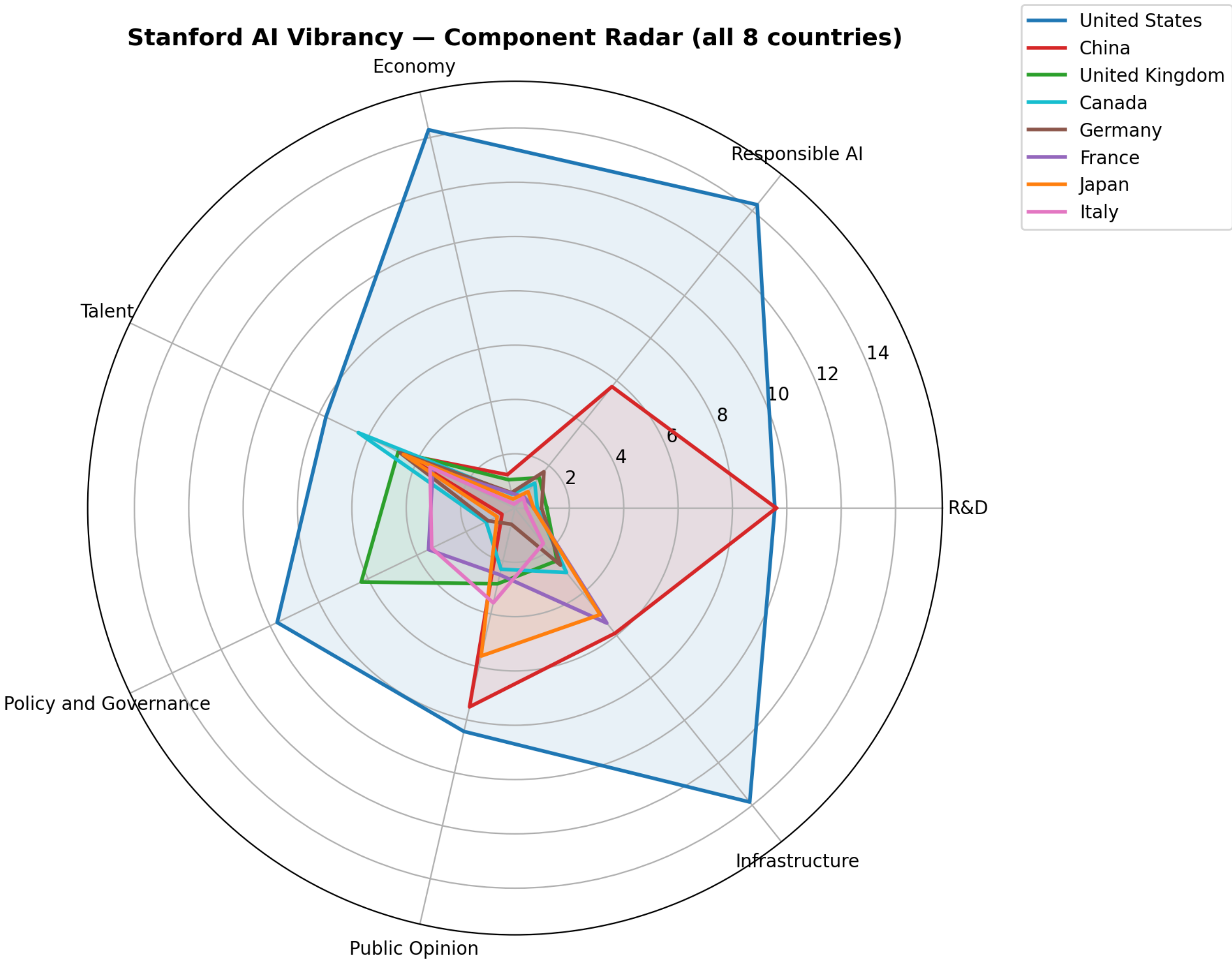
This framework explores whether AI ecosystem maturity influences regulatory rhetoric, and whether that rhetoric reflects broader economic and ideological preferences.

Preliminary Results

Stanford AI Vibrancy Tool — G7 + China

Cross-country comparison of AI vibrancy scores (Total Score and 7 sub-dimensions: R&D, Responsible AI, Economy, Talent, Policy and Governance, Public Opinion, Infrastructure).





Key observations:

- The United States leads overall (Total Score ≈ 77.85), driven primarily by Responsible AI, Economy, and Infrastructure.
- China ranks second (≈ 35.10) — its R&D score is essentially tied with the U.S., but it lags substantially in Responsible AI, Economy, and Policy & Governance.
- The remaining G7 countries cluster between 12 and 21, with no single G7 member dominating.
- The component radar chart highlights how each country's AI vibrancy profile differs across R&D, Responsible AI, Economy, Talent, Policy and Governance, Public Opinion, and Infrastructure.

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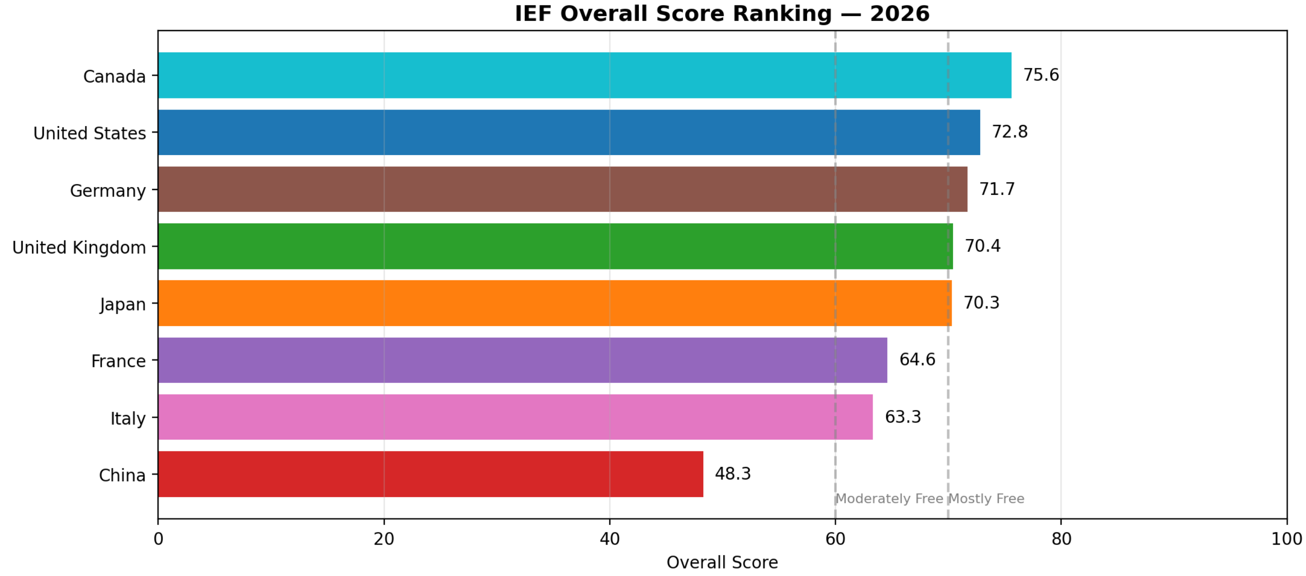
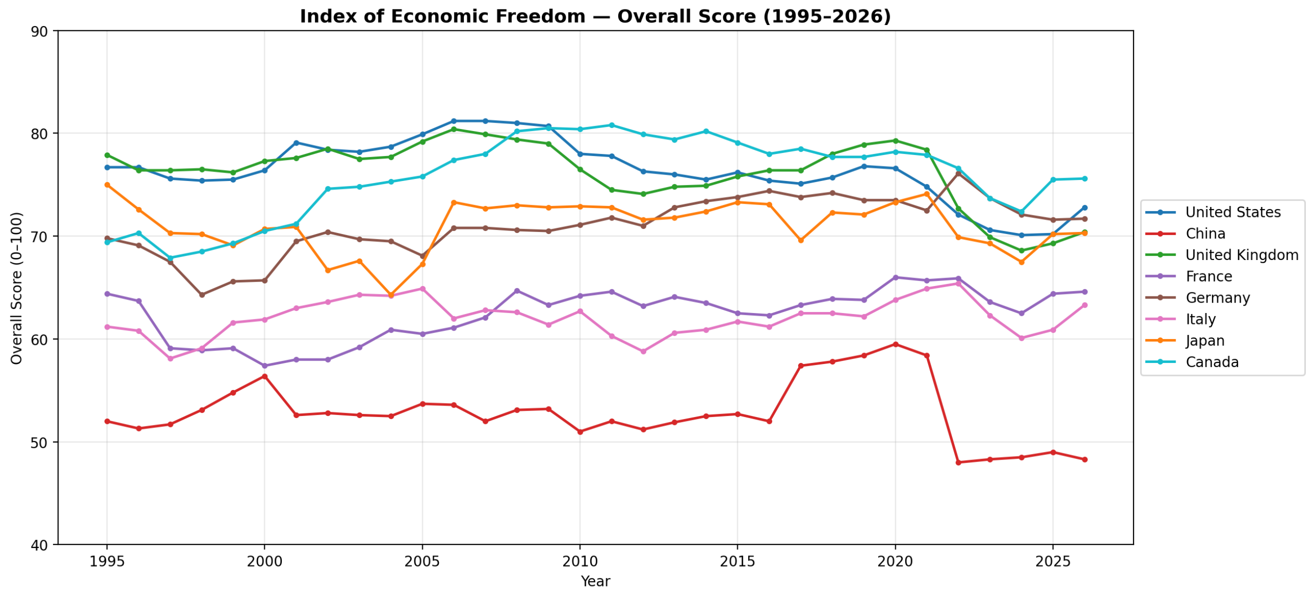
Index of Economic Freedom — G7 + China

Heritage Foundation Economic Freedom scores for the same eight countries, used as the political-economic context for the framing analysis.





Key observations:

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- G7 economies generally cluster in the "Mostly Free" / "Moderately Free" range.
- China sits noticeably below the G7, providing a natural contrast for testing whether economic freedom levels relate to AI regulatory framing.
- The yearly trend chart shows how economic freedom scores change over time, providing temporal context for the cross-sectional comparison.

Descriptive Statistics — Key Numeric Variables

To address the feedback, the project now explicitly reports standard descriptive statistics for the main numeric variables. The tables below report count, mean, median, standard deviation, minimum, and maximum for the Heritage Index of Economic Freedom scores and Stanford AI Vibrancy metrics.

Index of Economic Freedom

Variable	count	mean	median	stdev	min	max

Variable	count	mean	median	stdev	min	max
Overall Score	256	68.48	70.3	8.49	48.0	81.2
Property Rights	256	77.09	85.95	20.04	20.0	96.2
Government Integrity	256	70.47	76.0	17.9	22.0	95.1
Judicial Effectiveness	80	76.22	76.8	14.36	37.4	97.9
Tax Burden	256	61.41	62.35	10.62	33.2	80.0
Government Spending	256	43.14	43.95	22.77	0.0	95.9
Fiscal Health	80	48.77	54.55	30.69	0.0	92.9
Business Freedom	256	79.47	83.5	11.59	43.1	100.0
Labor Freedom	176	67.27	68.05	14.97	39.9	98.5
Monetary Freedom	256	81.03	81.5	5.96	61.5	94.3
Trade Freedom	256	79.42	80.8	9.94	20.0	88.8
Investment Freedom	256	66.19	70.0	18.3	20.0	90.0
Financial Freedom	256	63.12	70.0	17.72	20.0	90.0

Stanford AI Vibrancy

Variable	count	mean	median	stdev	min	max
Total Score	8	26.04	17.16	22.12	12.35	77.85
R&D	8	2.99	0.9	4.08	0.41	9.61
Responsible AI	8	3.26	1.3	4.77	0.49	14.29
Economy	8	2.35	0.57	4.84	0.15	14.29
Talent	8	4.94	4.63	1.46	3.35	7.72
Policy and Governance	8	3.31	2.29	3.24	0.53	9.71
Public Opinion	8	4.18	3.22	2.73	0.61	8.44
Infrastructure	8	5.01	4.03	3.89	1.67	13.85

The CSV versions are saved in `outputs/descriptive_statistics_ief.csv`, `outputs/descriptive_statistics_ai_vibrancy.csv`, and `outputs/descriptive_statistics_key_numeric_variables.csv`.

DeepSeek Corrected Innovation vs. Risk Framing

The corrected DeepSeek sentence-level classifier labels each policy sentence as innovation-oriented, risk-oriented, mixed, neutral, or unsure. Innovation-oriented sentences emphasize adoption, investment, productivity, competitiveness, regulatory flexibility, and deployment. Risk-oriented sentences emphasize safety, accountability, privacy, fairness, rights protection, regulatory obligations, and precaution.

Representative corrected label examples:

Country/entity	**Corrected label**	**Sentence excerpt**
Canada	Innovation-oriented	"AI can unlock capabilities beyond human limits, opening doors to new ways of working and operating."
Canada	Risk-oriented	"For greater data security and privacy, CANChat ensures that all data is safeguarded and stored in Canada, and that prompts are not used to train its AI."
European Union	Mixed	"122 The ‘ecosystem of trust’ focuses on measures to ensure that AI is developed in an ethical manner; the ‘ecosystem of excellence’ focuses on measures to promote responsible investment, innovation and implementation of AI."
France	Neutral	"Members include Ed Tech and universities such as Rennes, Haute-Alsace, Paris-Est Créteil, Bordeaux Montaigne, Nîmes, Montpellier, and the National Conservatory of Arts and Crafts."

The current corrected full run contains **3,730 sentences** across nine countries/entities, including the European Union for descriptive framing analysis. The strongest positive mean fram☰ing scores are France (`0.577`), the United Kingdom (`0.575`), and China (`0.598`). Canada (`0.088`) and the United States (`0.173`) show the lowest mean framing scores in the corrected summary.

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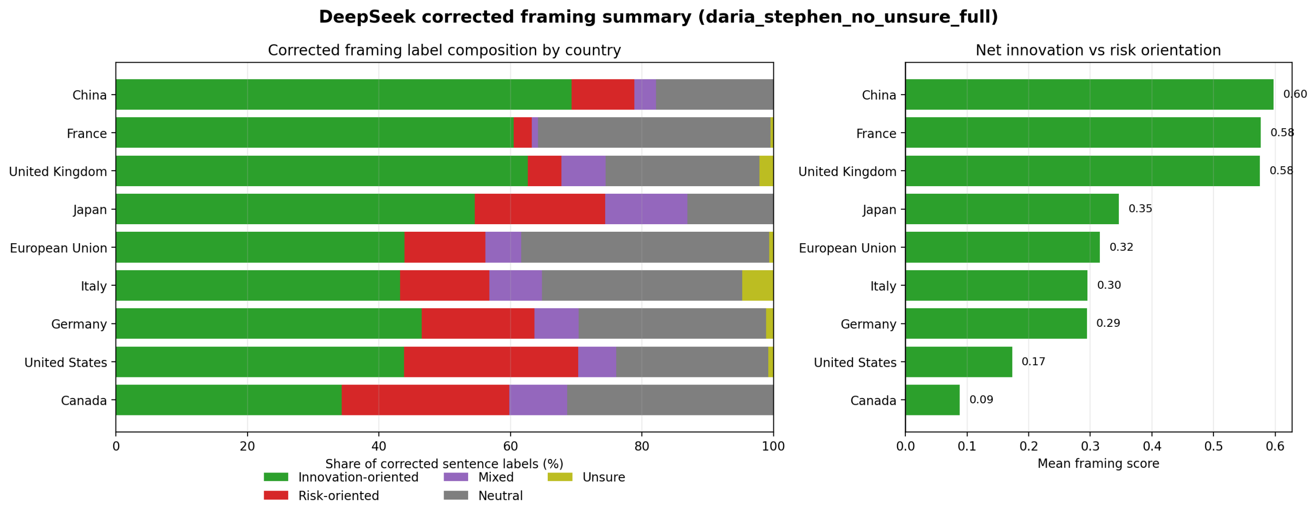


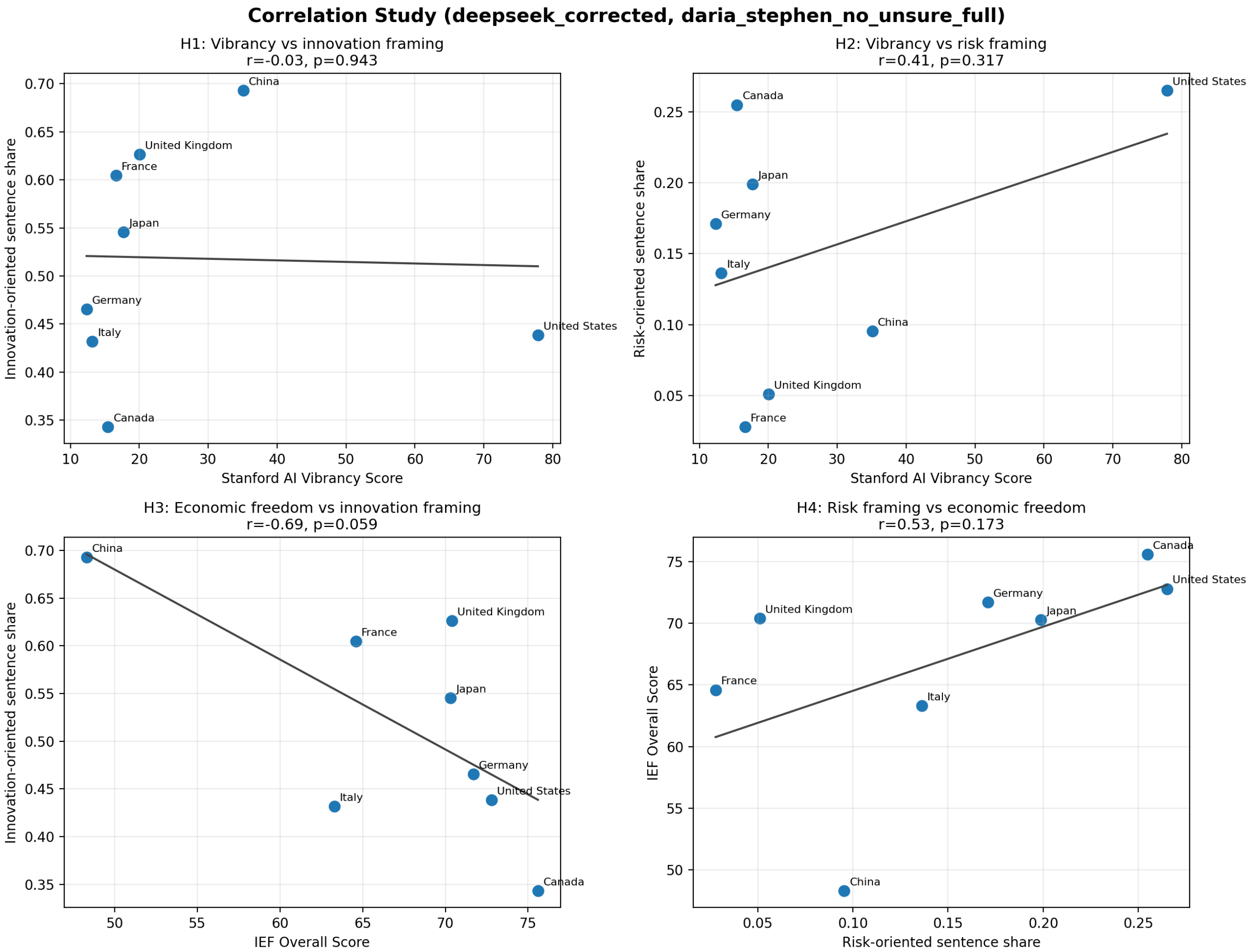
Outputs are saved in `outputs/deepseek_ai_framing_sentence_labels_raw_daria_stephen_no_unsure_full.csv`, `outputs/deepseek_ai_framing_sentence_labels_corrected_daria_stephen_no_unsure_full.csv`, `outputs/deepseek_ai_framing_summary_corrected_daria_stephen_no_unsure_full.csv`

, `outputs/deepseek_ai_framing_gold_metrics_daria_stephen_no_unsure_full.csv`, and `outputs/deepseek_ai_framing_gold_confusion_daria_stephen_no_unsure_full.csv`.

These corrected outputs provide the empirical anchor for H1-H4 by translating sentence-level regulatory discourse into country/entity-level innovation and risk framing measures.

Correlation Study — Vibrancy, Economic Freedom, and Framing

The correlation study combines Stanford AI Vibrancy scores, 2026 Heritage Index of Economic Freedom scores, and corrected DeepSeek framing results for the same G7 + China country set. The main hypothesis tests use `Innovation Share` and `Risk Share`; robustness checks use `Mean Framing Score` and the log smoothed innovation-to-risk ratio.



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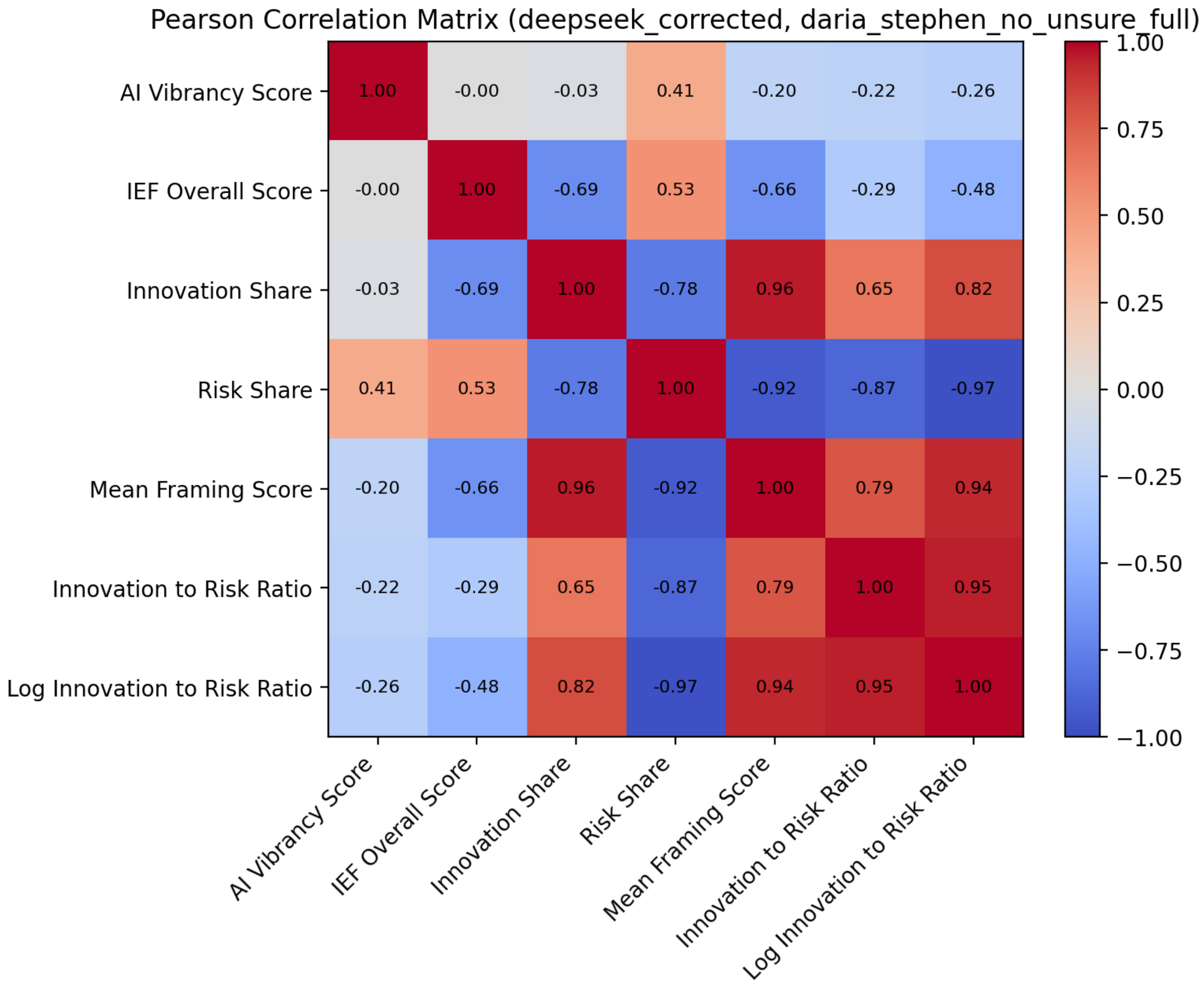

Current exploratory results using corrected DeepSeek framing:

- H1 expects AI Vibrancy to be positively associated with `Innovation Share`, but the current Pearson correlation is near zero and slightly negative (`r = -0.030`, `p = 0.9429`). Directional support is not observed.
- H2 expects lower AI Vibrancy to be associated with higher `Risk Share`, but the observed AI Vibrancy to Risk Share correlation is positive (`r = 0.407`, `p = 0.3172`). Directional support is not observed.
- H3 expects higher IEF scores to be associated with higher `Innovation Share`, but the observed relationship is negative (`r = -0.689`, `p = 0.0589`). Directional support is not observed.
- H4 expects `Risk Share` to be negatively associated with IEF, but the observed relationship is positive (`r = 0.534`, `p = 0.1728`). Directional support is not observed.
- Robustness checks using the log innovation-to-risk ratio also do not support the expected positive relationships with AI Vibrancy (`r = -0.264`, `p = 0.5280`) or IEF (`r = -0.478`, `p = 0.2312`).
- The combined model predicting log innovation-to-risk ratio from AI Vibrancy and IEF has `R2 = 0.298`, but with only eight observations it should be interpreted as a descriptive diagnostic.
- The mediation diagnostic is retained as a mechanism check, not a formal causal test.

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Key outputs are saved in `outputs/correlation_study_dataset_deepseek_corrected_daria_stephen_no_unsure_full.csv`, outputs/correlation_study_correlations_deepseek_corrected_daria_stephen_no_unsure_full.csv, outputs/correlation_study_regression_models_deepseek_corrected_daria_stephen_no_unsure_full.csv, and outputs/correlation_study_mediation_deepseek_corrected_daria_stephen_no_unsure_full.csv.

Folder Structure

```
AI-Policy/
├── data/                                    # Source data and
processed intermediate datasets
│   ├── AI vibrancy tool screen shot/        # Stanford AI Vibrancy
source screenshots and cleaned country scores
│   │   ├── ai_country_scores.csv            # Main AI Vibrancy
country-level score table used in correlations
│   │   ├── ai_country_scores_long.csv       # Long-format version for
plotting and diagnostics
│   │   └── *.png                            # Source screenshots
retained for traceability
│   ├── Human expert framing labels/         # Human annotation
materials for calibrating LLM sentence labels
│   │   ├── collected_data/                  # Raw collected human
annotation files
│   │   ├── data_colletion_tool/             # Annotation interface
and gold-label codebook materials
│   │   └── processed_data/                  # Daria + Stephen
agreement sets and agreement summaries
│   ├── Index of Economic Freedom/           # Raw Heritage Foundation
IEF country files
│   │   ├── G7+China All Data                # Combined source file
for the eight-country IEF panel
│   │   └── country-level IEF files          # Individual country
source files
│   ├── number/                              # Numeric datasets used
for AI development and economic analysis
│   │   ├── g7_china_datasets.csv            # Stanford AI Index files
containing enough G7 + China coverage
│   │   ├── ief_g7_china_panel.csv           # Clean 1995-2026 IEF
panel used in the correlation study
│   │   ├── ai_dev_index_g7_china.csv        # Derived AI development
index dataset
│   │   └── PUBLIC DATA_ 2026 AI INDEX REPORT/ # Stanford AI Index
public CSV source data
│   └── pdf/
│       └── AI Policy/                       # National AI strategy /
AI policy document corpus
│           ├── *_National_AI_Strategy*.pdf    # G7 + China source
policy PDFs
│           ├── EU_AI_Strategy.pdf           # EU benchmark policy
```

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```
document for descriptive framing analysis
│           ├── 国务院关于深入实施“人工智能+”行动的意见.pdf
│           ├── _extracted/                     # Extracted English text
and document-level diagnostics
│           │   ├── *_analysis_english.txt     # English analysis corpus
by country/entity
│           │   ├── document_sources.csv       # Mapping from text files
to source documents
│           │   └── document_stats.csv         # Word/sentence/document
diagnostics
│           └── _translated_nllb/              # NLLB translations for
Chinese-heavy texts
│               ├── China.txt
│               └── translation_stats.csv
│
├── notebooks/                                 # Reproducible analysis
notebooks
│   ├── AI Development/
│   │   ├── select_g7_china_data.ipynb         # Finds Stanford AI Index
CSVs with G7 + China coverage
│   │   ├── ai_vibrancy_tool.ipynb             # Cleans, summarizes, and
visualizes Stanford AI Vibrancy scores
│   │   └── ai_dev_index_v2.ipynb              # Builds an exploratory
composite AI development index
│   ├── Economic Freedom/
│   │   └── ief_g7_china_analysis.ipynb        # Builds the IEF panel
and saves IEF visualizations
│   ├── Text Analysis/
│   │   ├── national_ai_strategy_nllb_translation.ipynb
│   │   │                                      # Extracts/translates
policy text into the English analysis corpus
│   │   ├── human_expert_framing_labels_analysis.ipynb
│   │   │                                      # Processes human
annotation agreement and gold labels
│   │   └── LLM_based_text_framing_analysis.ipynb
│   │                                          # Runs DeepSeek sentence
labeling, human correction, and framing plots
│   └── Correlation Study/
│       └── correlation_study.ipynb            # Merges AI Vibrancy,
IEF, and corrected framing for H1-H4 tests
│
├── src/                                       # Shared Python helper
modules used by notebooks
│   ├── common.py                              # Country lists, colors,
document mappings, repo-root helper
│   ├── data_utils.py                          # CSV loading, country
extraction, markdown summary helpers
│   ├── ai_index.py                            # Score normalization and
weighted composite index helpers
│   ├── stats.py                               # Descriptive statistics
helper
│   ├── text_utils.py                          # PDF extraction,
sentence splitting, translation utilities
```

```
│   ├── human_framing.py                        # Human annotation
loading and agreement processing
│   └── llm_framing.py                          # DeepSeek/Ollama prompt,
parsing, correction, and summary logic
│
├── outputs/                                    # Generated tables,
figures, and report artifacts
│   ├── stanford_AI_Vibrancy.png                # README figure: AI
Vibrancy total score ranking
│   ├── stanford_AI_Vibrancy_component_radar.png
│   │                                           # README figure: AI
Vibrancy component radar
│   ├── IEF_score.png                           # README figure: latest
IEF ranking
│   ├── IEF_score_by_year.png                   # README figure: IEF
score trajectories over time
│   ├──
deepseek_ai_framing_sentence_labels_raw_daria_stephen_no_unsure_full.csv
│   │                                           # Raw DeepSeek sentence-
level labels before human override
│   ├──
deepseek_ai_framing_sentence_labels_corrected_daria_stephen_no_unsure_fu
ll.csv
│   │                                           # Final sentence labels
after exact human-gold correction
│   ├──
deepseek_ai_framing_summary_corrected_daria_stephen_no_unsure_full.csv
│   │                                           # Country/entity-level
corrected framing summary
│   ├──
deepseek_ai_framing_gold_metrics_daria_stephen_no_unsure_full.csv
│   │                                           # Human-gold evaluation
metrics for raw and corrected labels
│   ├──
deepseek_ai_framing_gold_confusion_daria_stephen_no_unsure_full.csv
│   │                                           # Raw model confusion
table against usable human-gold labels
│   ├──
deepseek_ai_framing_country_summary_corrected_daria_stephen_no_unsure_fu
ll.png
│   │                                           # README figure:
corrected label composition and mean framing score
│   ├──
correlation_study_dataset_deepseek_corrected_daria_stephen_no_unsure_ful
l.csv
│   │                                           # Merged G7 + China
dataset for correlation analysis
│   ├──
correlation_study_correlations_deepseek_corrected_daria_stephen_no_unsur
e_full.csv
│   │                                           # Pearson/Spearman H1-H4
correlation table
│   ├──
```

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```
correlation_study_regression_models_deepseek_corrected_daria_stephen_no_
unsure_full.csv
|   |                                               # OLS model coefficients
and model diagnostics
|   ├──
correlation_study_mediation_deepseek_corrected_daria_stephen_no_unsure_f
ull.csv
|   |                                               # Exploratory mediation
diagnostic
|   ├──
correlation_study_scatter_deepseek_corrected_daria_stephen_no_unsure_ful
l.png
|   ├──
correlation_study_matrix_deepseek_corrected_daria_stephen_no_unsure_full
.png
|   └── AI_policy_methodology_conceptual_framework.docx
|                                                   # Word methodology
document generated from current workflow
|
├── tests/                                          # Lightweight regression
tests and legacy test scaffolding
├── PolicyBrief.md                                  # Policy-facing written
summary draft
├── README.md                                       # Project overview,
methodology, results, and file guide
├── README.pdf                                      # PDF export of the
README/report
└── LICENSE
```

Key Structure Notes

`data/` contains both raw evidence and processed intermediate data. The most important reproducibility inputs are `ai_country_scores.csv` for AI Vibrancy, `ief_g7_china_panel.csv` for economic freedom, the AI policy PDFs under `data/pdf/AI Policy/`, and the human agreement files under `data/Human expert framing labels/processed_data/`.

`notebooks/` is organized by analytical stage. Run the AI Development and Economic Freedom notebooks to regenerate numeric predictors and README figures; run the Text Analysis notebooks to regenerate translated/extracted text, human-gold calibration files, and DeepSeek framing outputs; run `correlation_study.ipynb` last to rebuild the merged H1-H4 dataset, correlation tables, models, and final correlation figures.

`src/` holds shared helper code now imported directly from the flat `src` directory. The most important modules are `llm_framing.py` for local DeepSeek sentence classification and correction, `human_framing.py` for processing expert annotations, and `data_utils.py` / `common.py` for cross-notebook data handling.

`outputs/` contains generated artifacts rather than hand-edited source data. The DeepSeek files preserve the raw model labels, corrected sentence labels, human-gold diagnostics, and country-level framing summaries. The correlation files preserve the exact merged dataset and statistical outputs used in the README.